

C73 A1

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1 Q1

1.1 a: Does this algorithm minimize the sum of empty spaces on the shelves used? Justify your answer.

Yes, the statement is true. We prove the state follow the structure of the “greedy-stays-ahead” proof mentioned in note.

Consider a sequence of n books, let b_i represents the book, f_{b_i} represents the shelf which b_i in and $f_{b_i}^*$ be the optimal shelf for b_i , d_i represents the depth of book i , for $1 \leq i \leq n$.

Lemma: For every i , $1 \leq i \leq n$, $f_{b_i} \leq f_{b_i}^*$.

PROOF. By induction on i . The basis $i = 1$ is obvious by the algorithm. The first book has only one placement method which is absolutely the minimum.

For the induction step, suppose that $f_{b_i} \leq f_{b_i}^*$, prove that $f_{b_{i+1}} \leq f_{b_{i+1}}^*$. Suppose for contradiction, $f_{b_{i+1}} > f_{b_{i+1}}^*$, we can get (*) $f_{b_i} \leq f_{b_i}^* \leq f_{b_{i+1}}^* < f_{b_{i+1}}$. Then we consider two case:

CASE 1: there are t books in last shelf, which means $f_{b_{i-t+1}} = f_{b_{i-t+2}} = \dots = f_{b_{i+1}} = k$ and $\sum_{a=i-t+1}^{a=i+1} d_a \leq L$. In this way, we get $f_{b_i} \leq f_{b_i}^* \leq f_{b_{i+1}}^* = f_{b_{i+1}}$ which contradict (*).

CASE 2: $f_{b_{i-t+1}} = f_{b_{i-t+2}} = \dots = f_{b_i} = k$ and $\sum_{a=i-t+1}^{a=i+1} d_a > L$. Then we get $f_{b_{i+1}} = k + 1$. Since (*), we can get $f_{b_{i+1}}^* = k$, then for all book on k shelf, which means that the empty space for k shelf is less than 0. It is impossible.

Then $f_{b_i} \leq f_{b_i}^*$. QED.

Using this lemma, we can prove that the greedy algorithm is correct. Since we always get the minimum number of shelf, which means we place the book as tightly as possible. Assume, the placement according to algorithm can't give the optimal minimum sum of empty space, there must be another way to arrange. According to the lemma, there is impossible produce more empty space with less number of shelf. Thus, the statement holds.

1.2 b: Does this algorithm minimize the maximum empty space on any shelf used? Justify your answer.

No, the algorithm doesn't necessarily minimize the maximum empty space. The algorithm fill each shelf as much as possible before moving to next one but possibly leave one shelf with more empty space than others. Take the same counter example in Part a. $d_1 = 2, d_2 = 8, d_3 = 1$ with $L = 10$. In this example the first shelf will leave 0 space and second leave 9 empty space, so the maximum is 9. By rearranging the book, d_2 and d_3 in same shelf and there only 8 left in first and 1 left in two, the maximum is 8 .

2 Q2

2.1 a: True or false? For every set of intervals I and every minimum cardinality cover C of I , no two intervals in C intersect. Justify your answer.

False. It is unnecessary to make every two interval have no intersect. For example, $I = \{ [1, 3], [2, 5], [4, 7], [3, 4] \}$.

1. $C = \{ [1, 3], [3, 4], [4, 7] \}$ which has 3 interval but no intersect with each other.

2. $C = \{ [1, 3], [2, 5], [4, 7] \}$ which has 3 interval but have intersects.

Two kind of C are all satisfied with the requirement .

2.2 Describe an $O(n^2)$ greedy algorithm that, given a set I of n intervals, returns a minimum cardinality cover of I . Prove the correctness of your algorithm and analyze its running time.

b. Algorithm. The algorithm is as follows:

Algorithm 1 COVER($\mathcal{I}$)

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1:  $\mathcal{L} :=$  set of intervals in  $\mathcal{I}$  sorted by increasing finish time (if same just follow
   the recognition order)
2:  $\mathcal{A} := \emptyset$ 
3: while  $\mathcal{L} \neq \emptyset$  do
4:    $I :=$  first interval in  $\mathcal{L}$ 
5:   if  $I$  does not intersect any interval in  $\mathcal{A}$  then
6:      $I' :=$  interval in  $\mathcal{L}$  that intersects  $I$  and has maximum finish time
       (including itself).
7:      $\mathcal{A} := \mathcal{A} \cup \{I'\}$ 
8:   end if
9:    $\mathcal{L} := \mathcal{L} - \{I\}$ 
10: end while
11: return  $\mathcal{A}$ 

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Running time: Sorting the n interval in $\mathcal{I}$ by finish time cost $O(n \log n)$ time. The loop terminates after at most n iterations. Determine whether I intersect any interval in $\mathcal{A}$ (line 5) and find the interval I' (line 6) can each be done in $O(n)$ time. Thus, each iteration takes $O(n)$ and the entire algorithm takes $O(n^2)$.

Correctness:

First we notice that the loop will terminates, since in each iteration one interval will be removed from the $\mathcal{L}$ (line 9). Thus, after n iteration, the set will become empty.

To prove the output we get is optimal answer, I use the “promising set” proof mentioned in note.

Lemma: For every iteration i of the loop $\mathcal{A}$ is a cover of set of intervals removed from $\mathcal{L}$ and $\mathcal{A}$ is a subset of one optimal cover of $\mathcal{I}$.

PROOF: By the induction on i . Let L_i and A_i be the L and A at the end of i iteration. The basis $i = 0$ is obvious since $A_0 = \emptyset$ and nothing has been removed from L .

I.H. : Suppose that the lemma holds after i iteration, and A_i is a subset of some optimal set A_i^* . We will prove that (1) the A_{i+1} is a cover of the set of intervals in $\mathcal{I} - L_{i+1}$ and (2) A_i is a subset of some optimal set A_{i+1}^* of $\mathcal{I}$.

We can easily get that (1) is true by IH for (1) and the fact that the interval I deleted from L in iteration $i+1$ either intersect the interval which is already in A_i or intersects the interval I' added to A in iteration $i+1$. For (2), if the condition in Line 5 is false, then $A_i^* = A_{i+1}^*$ and we are done. Thus, we assume

that interval $I' = [s', f']$ is added to A and interval $I = [s, f]$ is removed in iteration $i+1$. We should consider two case:

CASE 1: $I' \in A_i^*$. Then we are done.

CASE 2: $I' \notin A_i^*$. Then there is a interval $I'' = [s'', f'']$, in A_i^* and intersects I . In this case, we replace the I'' by I' in A_i^* . We can claim that (a) A_{i+1}^* is optimal cover of $\mathcal{I}$ that contains A_{i+1} .

To prove this, note that $I'' \notin A_i$. By induction, $A_i \subseteq A_i^*$ and $I'' \notin A_i$. Thus, $A_{i+1} = A_i \cup \{I'\} \subseteq (A_i^* - \{I''\}) \cup \{I'\} = A_{i+1}^*$. A_{i+1}^* contains A_{i+1} and the same number of intervals as A_i^* . To prove (*) that A_{i+1}^* covers $\mathcal{I}$, it suffices to show (b) If $J \in \mathcal{I}$ intersects I'' , it also intersects some interval in A_{i+1}^* .

To prove (b), let $J = [s_J, f_J]$ be any interval in $\mathcal{I}$ that intersects I'' . If it also intersects I' , we are done since $I' \in A_{i+1}^*$. Suppose J does not intersect I' . Therefore, either (i) $f' < s_J$ or (ii) $f_J < s'$.

CASE 1: $f'' \leq f' < s_J$. This contradicts I'' intersects J , so this case is impossible.

CASE 2: $f_J < s' \leq f$. By induction, A_i covers all intervals deleted from $\mathcal{L}_i$, and since $f_J < f$, J is covered by A_i . We have shown $A_i \subseteq A_{i+1}^*$, so J is covered by A_{i+1}^* . This completes the proof of (b).

Now it is easy to show that the promising set lemma implied the algorithm return the wanted set. For the end condition of loop is that all interval in $\mathcal{I}$ is removed from $\mathcal{L}$. In this case the part 1 and 2 of lemma is raised so A must be optimal .

QED.